

Course plan for Bachelor's programme 2025/2026

Department of Advanced Computing Sciences, Maastricht University

Course code:	BCS1540
Course title:	Algorithmic Design
Bachelor:	Computer Science
Course coordinator/examiner (department):	David Mestel & Steven Chaplick
Teaching period:	1.5
Credits in ECTS:	4

1. General provisions of course assessment

Assessment at DACS is regulated by the *Education and Examination Regulations* (EER). Some of the main points therein are:

- Every course gets a single *final grade*, which is an integer from the range [1–10].
- A student passes the course, if and only if the *final grade* is 6.0 or higher.
- A resit of a course is provided during the same academic year. The details of the extent and form of the resit can vary from the regular exam, and is provided in Table 3 of this document.
- Student is allowed to inspect the grading of every assessment component of the course. The inspection of the final grade is regulated by EER: students may, within 10 working days after the result (grade) for the course has been made official, inspect their assessment, and the required standards based on which the assessment has been done.

2. Graded components of the course assessment

During the teaching and examination period of the course, the following components will be graded, and will count towards the final grade of the course:

- 30-minute written assignment under controlled conditions in Tutorial in week 3.
- Written exam

3. Calculation of the final grade

$\text{Max}(\text{exam grade}, 0.9 \cdot \text{exam grade} + 0.1 \cdot \text{assignment grade})$

4. Relation of the assessment to the learning outcomes of the course and to the Bachelor program

Intended learning outcomes (ILOs)	Teaching & learning activities (TLAs)	Assessment component	Programme qualifications
Knowledge and understanding: Students will be able to understand the main algorithmic paradigms and techniques. They are familiar with the notion of complexity classes.	- Lectures - Homework	- Exam - Assignment	1, 2
Applying knowledge and understanding: Students are able to propose an algorithm for solving a given problem.	- Tutorials - Homework	- Exam - Assignment	4, 7, 9
Making judgement: Students will show good design principles and adequate skills at reasoning about the correctness and complexity of algorithms.	-Tutorials -Homework	- Exam - Assignment	14
Communication: Students are able to explain how algorithms are to be included in program designs.	- Lectures - Tutorials	- Exam - Assignment	18
Learning skills: Students are able to learn sophisticated algorithms and analytical techniques, identifying and filling gaps in their knowledge via exercises and appropriate sources.	- Tutorials - Homework	- Exam - Assignment	19

Table 1: Course intended learning outcomes (ILOs), corresponding teaching and learning activities (TLAs, corresponding assessment components, and the covered qualifications of the whole Bachelor program)

5. Form of the assessment components and their grading

Assessment component	Form and extent (length)	Grading scheme/key	Assessor
Assignment	30-minute assignment under controlled conditions (closed-book, no electronics) in a tutorial	Graded similarly to exam	TA (junior lecturer)
Written exam	Closed book exam, 2 hours	Points are given for each correctly answered question	The examiners

Table 2: Assessment components, form of assessment, grading scheme, and assessor

For the resit, some assessment components may be assessed differently. The details are given in Table 3.

Examination component	Form of the resit
Resit	Identical structure to the regular exam. The final grade of the resit is computed as the maximum of: 10% x Assignment Grade + 90% x Resit Exam Grade 100% x Resit Exam Grade

Table 3: Resit forms of all examination elements